

Class 11: AlphaFold

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Custom Analysis of Resulting Models

```
library(bio3d)
pdb <- read.pdb("hivp_dimer_0d832/hivp_dimer_0d832_unrelaxed_rank_001_alphafold2_ptm_model_1.pdb")
```

```
Call: read.pdb(file = "hivp_dimer_0d832/hivp_dimer_0d832_unrelaxed_rank_001_alphafold2_ptm_model_1.pdb")
```

```
Total Models#: 1
```

```
Total Atoms#: 1440, XYZs#: 4320 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1440 (residues/Calpha atoms#: 176)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 0 (residues: 0)
```

```
Non-protein/nucleic resid values: [ none ]
```

```
Protein sequence:
```

```
MRDEIATTVFFVTRLVKKHNLKSKQIEDFAEKLMTILFETYRSHWHSDCPSKGQAFRCIR
INNQNKNKDPIERACAESKVDFFHLGLPKELTIWDPFEVCCRYGEKSHPFTVASFKGRWEE
WELYQQISYAVSRASSDDSSGTSCNEESCGKEPHVIPKVSNPFSIYQVENLKQPFQ
```

```
+ attr: atom, xyz, calpha, call
```

Make a vector of input PDB file names that we can read into R

```
pdb_files <- list.files("hivp_dimer_0d832/",  
  pattern = ".pdb",  
  full.names = TRUE)  
basename(pdb_files)
```

```
[1] "hivp_dimer_0d832_unrelaxed_rank_001_alphafold2_ptm_model_1_seed_000.pdb"  
[2] "hivp_dimer_0d832_unrelaxed_rank_002_alphafold2_ptm_model_2_seed_000.pdb"  
[3] "hivp_dimer_0d832_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb"  
[4] "hivp_dimer_0d832_unrelaxed_rank_004_alphafold2_ptm_model_4_seed_000.pdb"  
[5] "hivp_dimer_0d832_unrelaxed_rank_005_alphafold2_ptm_model_5_seed_000.pdb"
```

```
# Read all data from Models  
# and superpose/fit coords  
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

Reading PDB files:

```
hivp_dimer_0d832//hivp_dimer_0d832_unrelaxed_rank_001_alphafold2_ptm_model_1_seed_000.pdb  
hivp_dimer_0d832//hivp_dimer_0d832_unrelaxed_rank_002_alphafold2_ptm_model_2_seed_000.pdb  
hivp_dimer_0d832//hivp_dimer_0d832_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb  
hivp_dimer_0d832//hivp_dimer_0d832_unrelaxed_rank_004_alphafold2_ptm_model_4_seed_000.pdb  
hivp_dimer_0d832//hivp_dimer_0d832_unrelaxed_rank_005_alphafold2_ptm_model_5_seed_000.pdb  
.....
```

Extracting sequences

```
pdb/seq: 1   name: hivp_dimer_0d832//hivp_dimer_0d832_unrelaxed_rank_001_alphafold2_ptm_model_1_seed_000.pdb  
pdb/seq: 2   name: hivp_dimer_0d832//hivp_dimer_0d832_unrelaxed_rank_002_alphafold2_ptm_model_2_seed_000.pdb  
pdb/seq: 3   name: hivp_dimer_0d832//hivp_dimer_0d832_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb  
pdb/seq: 4   name: hivp_dimer_0d832//hivp_dimer_0d832_unrelaxed_rank_004_alphafold2_ptm_model_4_seed_000.pdb  
pdb/seq: 5   name: hivp_dimer_0d832//hivp_dimer_0d832_unrelaxed_rank_005_alphafold2_ptm_model_5_seed_000.pdb
```

```
pdbs
```

```
1 . . . . 50  
[Truncated_Name:1]hivp_dimer MRDEIATTVFFVTRLVKKHNKLSKQIEDFAEKLMTILFETYRSHWSDCP  
[Truncated_Name:2]hivp_dimer MRDEIATTVFFVTRLVKKHNKLSKQIEDFAEKLMTILFETYRSHWSDCP  
[Truncated_Name:3]hivp_dimer MRDEIATTVFFVTRLVKKHNKLSKQIEDFAEKLMTILFETYRSHWSDCP  
[Truncated_Name:4]hivp_dimer MRDEIATTVFFVTRLVKKHNKLSKQIEDFAEKLMTILFETYRSHWSDCP
```

```

[Truncated_Name:5]hivp_dimer MRDEIATTVFFVTRLVKKHKNLSKQIEDFAEKLMTILFETYRSHWSDCP
*****
1 . . . . 50

51 . . . . 100
[Truncated_Name:1]hivp_dimer SKGQAFRCIRINNNQNKDPIERACAESKVDFHGLPKELTIWDPFEVCC
[Truncated_Name:2]hivp_dimer SKGQAFRCIRINNNQNKDPIERACAESKVDFHGLPKELTIWDPFEVCC
[Truncated_Name:3]hivp_dimer SKGQAFRCIRINNNQNKDPIERACAESKVDFHGLPKELTIWDPFEVCC
[Truncated_Name:4]hivp_dimer SKGQAFRCIRINNNQNKDPIERACAESKVDFHGLPKELTIWDPFEVCC
[Truncated_Name:5]hivp_dimer SKGQAFRCIRINNNQNKDPIERACAESKVDFHGLPKELTIWDPFEVCC
*****
51 . . . . 100

101 . . . . 150
[Truncated_Name:1]hivp_dimer RYGEKSHPFTVASFKGRWEEWELYQQISYAVSRASSDDSSGTSCNEESCG
[Truncated_Name:2]hivp_dimer RYGEKSHPFTVASFKGRWEEWELYQQISYAVSRASSDDSSGTSCNEESCG
[Truncated_Name:3]hivp_dimer RYGEKSHPFTVASFKGRWEEWELYQQISYAVSRASSDDSSGTSCNEESCG
[Truncated_Name:4]hivp_dimer RYGEKSHPFTVASFKGRWEEWELYQQISYAVSRASSDDSSGTSCNEESCG
[Truncated_Name:5]hivp_dimer RYGEKSHPFTVASFKGRWEEWELYQQISYAVSRASSDDSSGTSCNEESCG
*****
101 . . . . 150

151 . . 176
[Truncated_Name:1]hivp_dimer KEPHVIPKVSNPFSIYQVENLKQPFQ
[Truncated_Name:2]hivp_dimer KEPHVIPKVSNPFSIYQVENLKQPFQ
[Truncated_Name:3]hivp_dimer KEPHVIPKVSNPFSIYQVENLKQPFQ
[Truncated_Name:4]hivp_dimer KEPHVIPKVSNPFSIYQVENLKQPFQ
[Truncated_Name:5]hivp_dimer KEPHVIPKVSNPFSIYQVENLKQPFQ
*****
151 . . 176

```

Call:

```
pdaln(files = pdb_files, fit = TRUE, exefile = "msa")
```

Class:

```
pdbs, fasta
```

Alignment dimensions:

```
5 sequence rows; 176 position columns (176 non-gap, 0 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

```
rd <- rmsd(pdbbs, fit=T)
```

Warning in rmsd(pdbbs, fit = T): No indices provided, using the 176 non NA positions

```
range(rd)
```

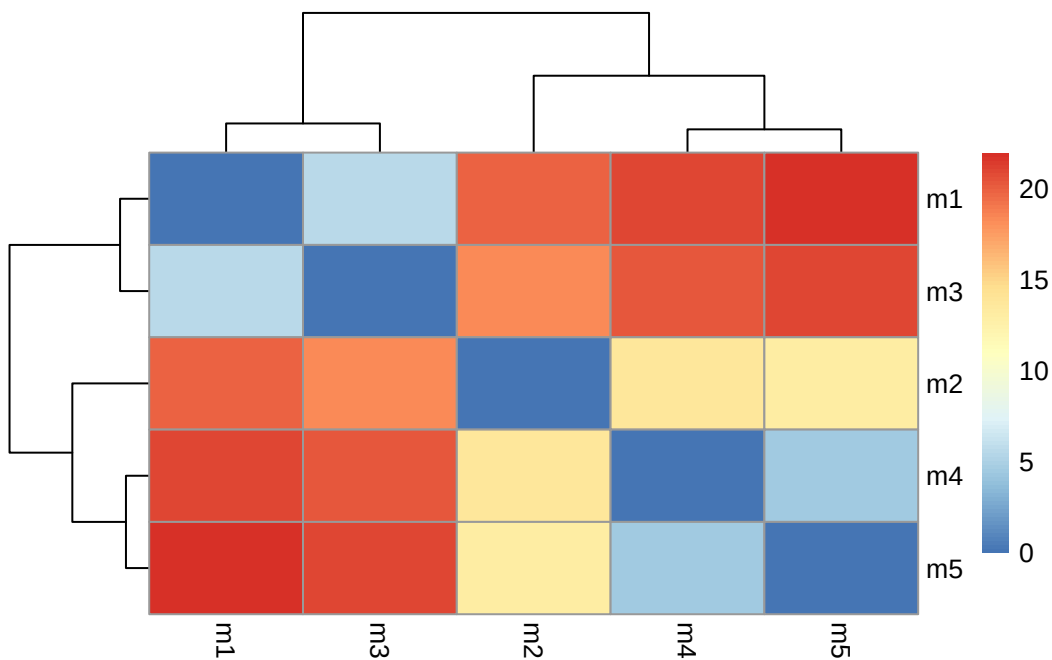
```
[1] 0.000 21.947
```

```
library(pheatmap)
```

```
colnames(rd) <- paste0("m",1:5)
```

```
rownames(rd) <- paste0("m",1:5)
```

```
pheatmap(rd)
```



The RMSD heatmap compares structural similarity between the five AlphaFold models. Additionally, the lower RMSD values indicate more similar structures while models with higher RMSD are more structurally different from the others.

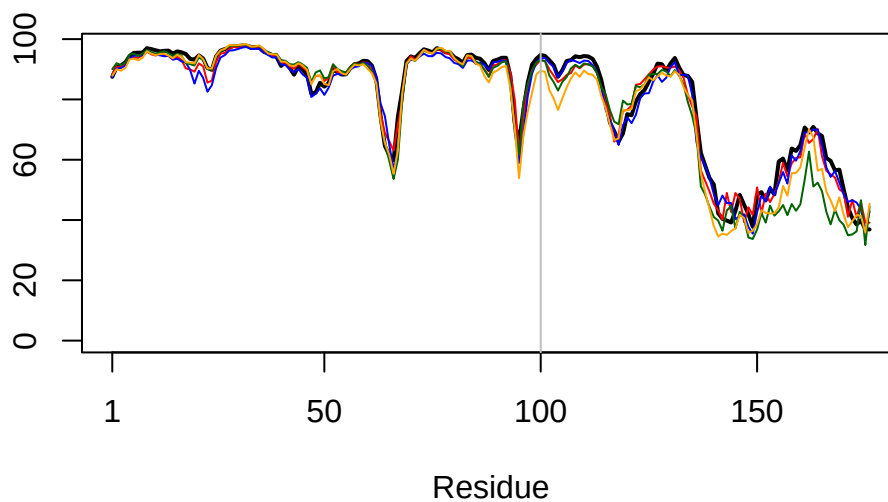
```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
plotb3(pdbb$b[1,], typ="l", lwd=2, sse=pdb)
```

Warning in plotb3(pdbb\$b[1,], typ = "l", lwd = 2, sse = pdb): Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'

```
points(pdbb$b[2,], typ="l", col="red")
points(pdbb$b[3,], typ="l", col="blue")
points(pdbb$b[4,], typ="l", col="darkgreen")
points(pdbb$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



The pLDDT plot shows AlphaFold confidence across residue positions. Moreover, the higher pLDDT values indicate higher confidence, while lower values suggest less reliable regions of the model.

```
core <- core.find(pdbb)
```

```
core size 175 of 176  vol = 59740.11
core size 174 of 176  vol = 49331.57
core size 173 of 176  vol = 46442.46
core size 172 of 176  vol = 44034.47
```

core size 171 of 176 vol = 41663.07
core size 170 of 176 vol = 39972.2
core size 169 of 176 vol = 37971.21
core size 168 of 176 vol = 36135.77
core size 167 of 176 vol = 34793.5
core size 166 of 176 vol = 33767.56
core size 165 of 176 vol = 32341.92
core size 164 of 176 vol = 31045.85
core size 163 of 176 vol = 29234.63
core size 162 of 176 vol = 27439.82
core size 161 of 176 vol = 25645.37
core size 160 of 176 vol = 23902.96
core size 159 of 176 vol = 21973.68
core size 158 of 176 vol = 20161.16
core size 157 of 176 vol = 18486.13
core size 156 of 176 vol = 17019.19
core size 155 of 176 vol = 15769.75
core size 154 of 176 vol = 14503.63
core size 153 of 176 vol = 13657.27
core size 152 of 176 vol = 13029.01
core size 151 of 176 vol = 12707.01
core size 150 of 176 vol = 12781.06
core size 149 of 176 vol = 12943.19
core size 148 of 176 vol = 12555.66
core size 147 of 176 vol = 12205.02
core size 146 of 176 vol = 12247.06
core size 145 of 176 vol = 11547.95
core size 144 of 176 vol = 10764.27
core size 143 of 176 vol = 9999.411
core size 142 of 176 vol = 9230.039
core size 141 of 176 vol = 8041.899
core size 140 of 176 vol = 6786.012
core size 139 of 176 vol = 5173.183
core size 138 of 176 vol = 3247.493
core size 137 of 176 vol = 911.866
core size 136 of 176 vol = 210.723
core size 135 of 176 vol = 162.178
core size 134 of 176 vol = 138.219
core size 133 of 176 vol = 118.938
core size 132 of 176 vol = 106.287
core size 131 of 176 vol = 95.029
core size 130 of 176 vol = 81.78
core size 129 of 176 vol = 73.211

```

core size 128 of 176 vol = 64.041
core size 127 of 176 vol = 56.801
core size 126 of 176 vol = 47.741
core size 125 of 176 vol = 39.084
core size 124 of 176 vol = 33.14
core size 123 of 176 vol = 28.014
core size 122 of 176 vol = 22.268
core size 121 of 176 vol = 17.934
core size 120 of 176 vol = 13.563
core size 119 of 176 vol = 9.068
core size 118 of 176 vol = 4.391
core size 117 of 176 vol = 1.548
core size 116 of 176 vol = 0.652
core size 115 of 176 vol = 0.523
core size 114 of 176 vol = 0.355
FINISHED: Min vol ( 0.5 ) reached

```

```
core.inds <- print(core, vol=0.5)
```

```
# 115 positions (cumulative volume <= 0.5 Angstrom^3)
```

```

  start end length
1      1  65      65
2     67 116      50

```

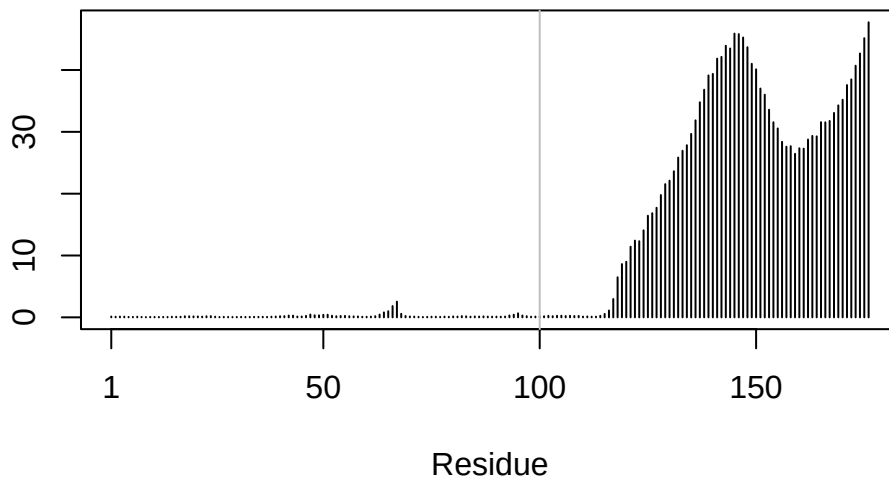
```
xyz <- pdbfit(pdb, core.inds, outpath="corefit_structures")
```

```
rf <- rmsf(xyz)
```

```
plotb3(rf, sse=pdb)
```

Warning in plotb3(rf, sse = pdb): Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'

```
abline(v=100, col="gray", ylab="RMSF")
```



The RMSF plot shows how much each residue varies across the fitted models as the higher RMSF values indicate more flexible or variable regions.

Predicted alignment error for domains

```
library(jsonlite)
# Listing of all PAE JSON files
results_dir <- "hivp_dimer_0d832/"
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)

pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"  "max_pae" "pae"     "ptm"
```



```
# Per-residue pLDDT scores
# same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 87.50 90.62 91.25 92.25 94.38 95.44
```

```
pae1$max_pae
```

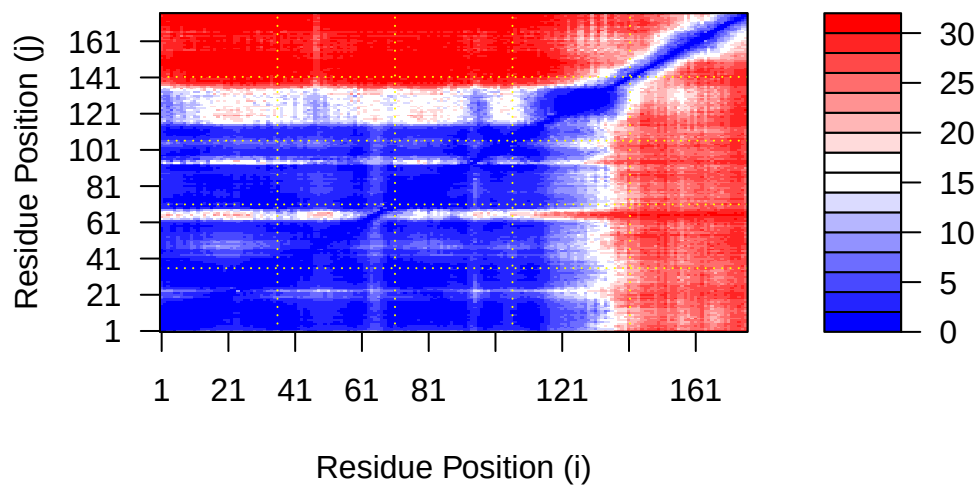
```
[1] 31.29688
```

```
pae5$max_pae
```

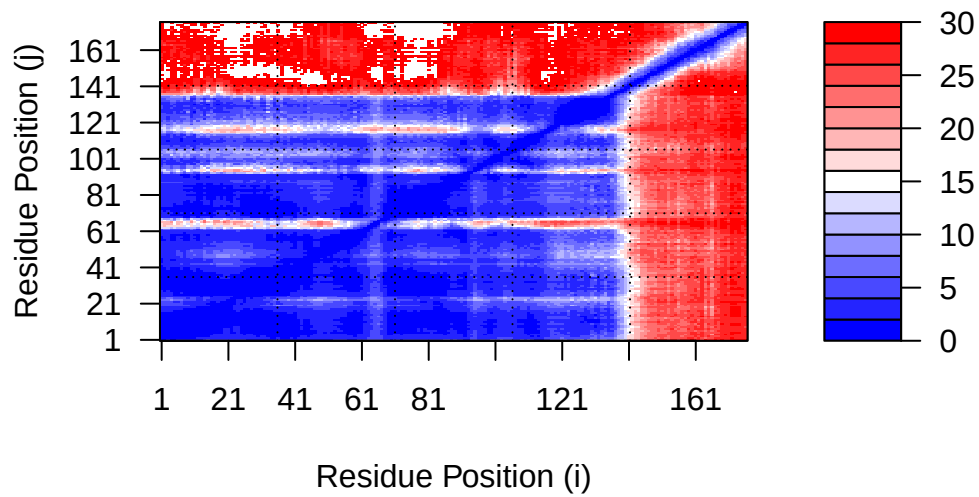
```
[1] 31.14062
```

The maximum PAE values compare prediction uncertainty between models. A lower max PAE indicates a more reliable model. Hence, the model 1 is more reliable than the model 5.

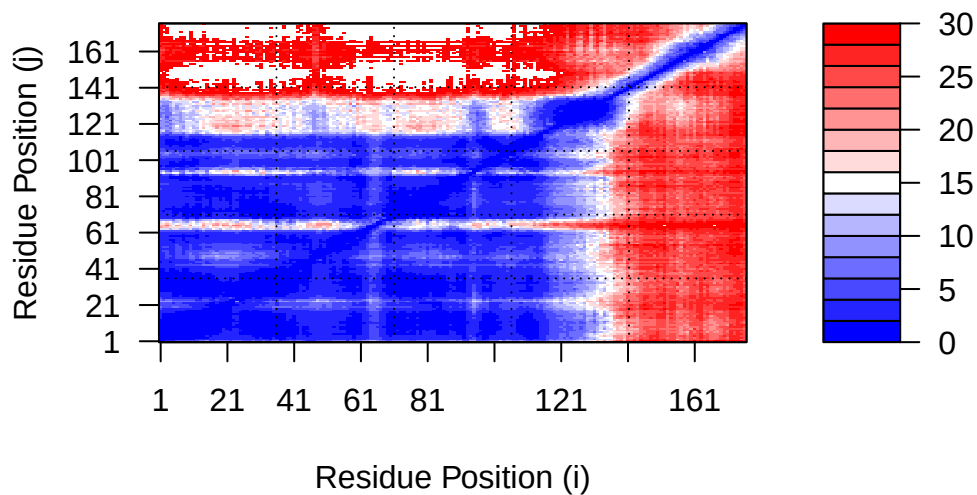
```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```



```
plot.dmat(pae5$pae,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30))
```



```
plot.dmat(pae1$pae,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30))
```



The PAE plots show predicted alignment error between residue pairs. The first Model 1 shows lower overall error than model 5. This suggests a more confident domain arrangement.

Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                      pattern=".a3m$",
                      full.names = TRUE)
aln_file
```

```
[1] "hivp_dimer_0d832//hivp_dimer_0d832.a3m"
```

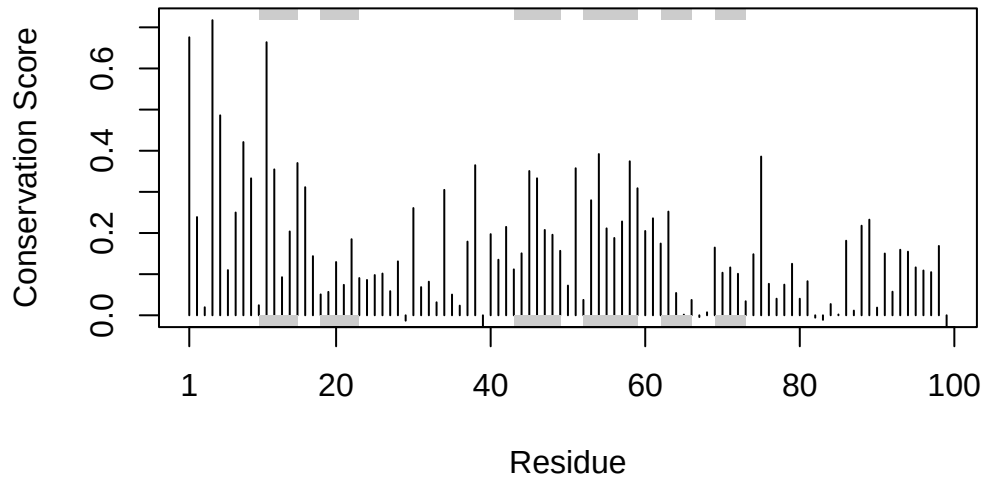
```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 2704 232
```

```
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



The conservation plot shows which residues are most conserved across the sequence alignment.

```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[145] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[163] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[181] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[199] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[217] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "
```

```

m1.pdb <- read.pdb(pdb_files[1])

# uses conservation scores for all residues in the model
sim2 <- rep(sim[1:99], 2)

# map residue conservation scores onto every atom
occ <- sim2[match(m1.pdb$atom$resno, sort(unique(m1.pdb$atom$resno)))]

# write new PDB file for Mol Star
write.pdb(m1.pdb, o = occ, file = "m1_conserv.pdb")

```

The original code produced a length mismatch error, so it was adjusted to map conservation scores to residue numbers.